



Automated Coin Counter – Project Design

PROCTECH 2SE3 – Systems Engineering Fundamentals

Winter 2026

The W Booth School of Engineering Practice and Technology

McMaster University, Hamilton, Ontario, Canada

Submitted By:

Luna Kim – 400521674

Majd Aburas – 400509985

Lyba Naseem – 400602518

James Giovinazzo – 400583359

Course Section:

C01, Group 9

Instructor:

Konstantinos Apostolou

Date of submission:

Sunday, March 8th, 2026

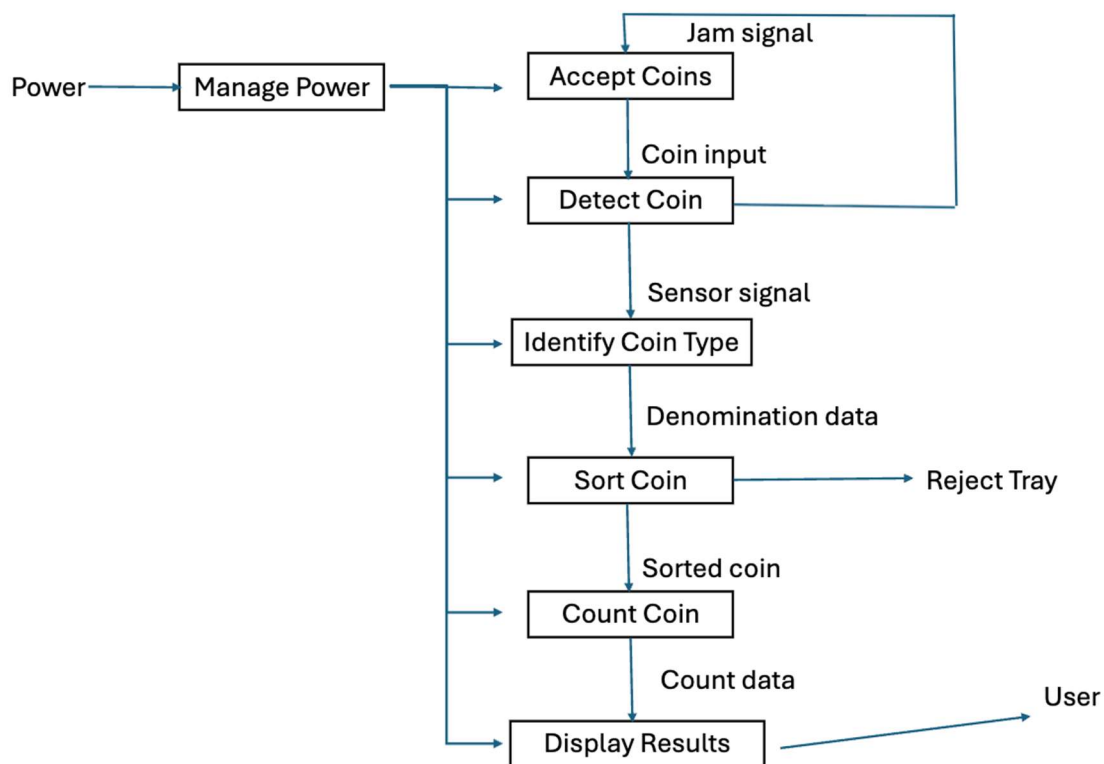
Table of Contents

1. Project Description	3
2. Functional Design.....	3
3. Morphological Chart	4
4. Design Alternatives.....	5
5. Design Selection	6
6. Selected Design Statement	7

1. Project Description

Automated coin sorting and counting system is designed to recognize, sort, and count mixed coins without requiring human involvement. It is used in commercial and financial environments where efficiency and accuracy are important. The system reduces manual handling and operational errors while maintaining safe and reliable operation. It continuously monitors operating conditions and responds in a controlled manner. It is also designed to operate safely during power outages and to make routine maintenance easier.

2. Functional Design



3. Morphological Chart

Function	Option 1	Option 2	Option 3
Coin Input Method	Coins are inserted manually into the machine.	Coins are inserted automatically using a feeder.	Coins are inserted through a guided slot.
Coin Detection	The system detects coins using an optical sensor.	The system detects coins using an IR sensor.	The system detects coins using a camera system.
Coin Identification	The system identifies coins by size.	The system identifies coins using image recognition.	The system identifies coins by weight.
Coin Transport	Coins move through the system using gravity.	Coins move using a rotating disk.	Coins move using a conveyor belt system.
Coin Sorting	Coins are separated using a rotating sorter.	Coins are separated using mechanical gates.	Coins are separated using motorized gates.
Coin Counting	Coins are counted using encoder.	Coins are counted using pulse signals.	Coins are counted using optical sensors
Reject Handling	Invalid coins are returned to the user.	Invalid coins go to a reject tray.	Invalid coins are stored in a reject container.
Result Display	Results are displayed using an LED display.	Results are displayed using an LCD screen.	Results are displayed using a touchscreen display.

4. Design Alternatives

Alternative 1: Ease of Use

This design focuses on minimizing user effort. Instead of manually inserting coins, it uses an automatic feeder so the operator can dump a batch of coins and walk away. We added a touchscreen display for an intuitive, modern interface. To prevent user frustration, invalid coins are returned directly to the user rather than sent to a reject bin. What makes this unique is its focus on a frictionless operator experience rather than optimized efficiency.

Alternative 2: Lowest Cost

Here, the goal is to keep the manufacturing costs as low as possible. We removed expensive actuators, relying entirely on manual coin insertion and gravity for transport. Detection and identification are handled by basic optical sensors that check the coin size, which is much cheaper than running a full camera system. Sorting uses simple mechanical gates. This alternative stands out because it relies heavily on passive physics and cheap, off-the-shelf electronics rather than complex control systems.

Alternative 3: Highest Accuracy

This option prioritizes strict precision to hit our 99.5% accuracy goal. Coins enter through a guided slot to ensure perfect alignment for the camera system, which identifies them using image recognition. Transport relies on a rotating disk to keep coins evenly spaced, sorting uses precise motorized gates, and counting is tracked by an encoder. Its uniqueness lies in using closed-loop, vision-based validation to practically eliminate false positives.

Alternative 4: Fast Processing

Optimized for high throughput, this design aims to easily meet the 60 coins-per-minute requirement. An automatic feeder keeps a constant stream going onto a fast conveyor

belt. To avoid computational bottlenecks from image processing, it uses rapid IR sensors and sorts coins physically by size using a continuous rotating sorter. It's unique because it prioritizes rapid physical sorting and fast sensor polling over complex, slower identification checks.

5. Design Selection

Objective weighing for normalized weight

Objective	Ease of Use	Low Cost	Accuracy	High Speed	Total Score	Weight (Normalized)
Ease of Use	—	1	0	0.5	1.5	0.25
Low Cost	0	—	0	0	0	0
Accuracy	1	1	—	1	3	0.5
High Speed	0.5	1	0	—	1.5	0.25
Total					6	1

Objective	Weight	Ease	Cost	Accuracy	Speed
Accuracy	0.5	7	4	10	8
Ease of Use	0.25	10	3	6	8
High Speed	0.25	6	2	7	10
Weighted Total	1	7.5	3.25	8.25	8.5

- **Accuracy (Weight 0.50):** It is assigned the highest weight because it is more important to all other criteria. When comparing Accuracy vs. Speed, Accuracy wins (a slow, correct machine is better than a fast, incorrect one). When comparing Accuracy vs. Ease of Use, Accuracy wins.

- High Speed (Weight 0.25): We assigned this higher importance than Cost, because the primary commercial goal of this machine is to replace manual counting. If it fails to speed up the process, it fails its main task.
- Ease of Use (Weight 0.25): We weighted this equally with Speed. In a retail environment, if the user interface is too complex, employees will avoid using the machine, rendering the system useless regardless of its speed.
- Low Cost (Weight 0.00): We weighted this lowest (or zero) because our design goal is "Professional Commercial Grade." In this specific project scope, we decided that performance reliability is more valuable to the customer than the lowest possible manufacturing price.

6. Selected Design Statement

The selected design for the automated coin counter is alternative 4: high speed processing, prioritizing efficient and fast handling of coins without compromising accurate counting and sorting.

In this design, coins are automatically inserted using a feeder, which keeps a constant stream going onto a fast conveyer belt. Coin detection is done using rapid IR sensors and a rotating sorting mechanism, which allow identification without any delays like in the camera system. Invalid coins are directed to a reject tray, showing final results to the user on an LCD display.

This design provides the user with an efficient and reliable operation, making it ideal for environments where large amounts of coins must be counted quickly.

Provide a clear statement of your selected, best design alternative. Ensure you mention all elements from the morphological chart included in this design solution.

This report was authored by ALL members of PROCTECH 2SE3 Group 9, with minimal assistance from Artificial Intelligence (Google's 'Gemini 3 Pro' language model, March 2026), for the sole purpose of polishing language and improving clarity. All ideas, data analysis, and conclusions presented in this report are entirely our own.